Exponential & Log Mirror

Explore & Reflect

Name _____ Date _____ Period _____

Open **Exponential & Log Mirror** on Mr. K's Classroom: Algebra 2 → Exponential & Logarithmic Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If $2^3 = 8$, what is $\log_2 8$?

2 Explore use the tool and record what you see

Matching points

Move the input slider and record both points.

Exponential form	Log form	Point on $y = b^x$	Point on $y = \log_b x$
$2^3 = 8$			
$2^{-1} = \frac{1}{2}$			
$3^2 = 9$			
$10^0 = 1$			

Finding e: $1 at 100% interest for 1 year

Compounded	yearly	twice	quarterly	monthly	daily	every minute
Value	$2					

3 Write the rule fill in each blank from what you observed

1. $\log_b y = x$ means _____.
2. The graphs of $y = b^x$ and $y = \log_b x$ are reflections over _____.
3. The domain of $y = b^x$ is _____ and its range is _____. For $y = \log_b x$ they _____.
4. Every graph $y = \log_b x$ passes through _____, because $b^0 = 1$.
5. As interest compounds more often, $(1 + \frac{1}{n})^n$ gets closer to _____ ≈ 2.718.

4 Check no tool: show your work

1. Evaluate $\log_5 125$.

2. Evaluate $\log_4 \frac{1}{16}$.

3. Rewrite $10^3 = 1000$ in log form.

4. Find the domain of $y = \log_2(x - 3)$.

5 Reflect

Why can't you take the log of a negative number?